Consistent sets of first-order sentences have enumerable models

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openlogic:first-order-logic/completeness:local-snapshot-20261013;
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Editorial scope. Statement and complete author proof are included verbatim as the span lines around ollabel thm:downward-ls of the pinned original source; no text is rewritten or paraphrased.

Allowed standard prerequisites (editorial).

- the completeness theorem's term-model construction
- the term set of a countable language and its cardinality
- finite and denumerable domains

Theorem 0.1. *If* Γ *is consistent then it has !!aenumerable model, i.e., it is satisfiable in !!astructure whose domain is either finite or !!denumerable.*

Proof. If Γ is consistent, the !!structure $\mathfrak{A}M$ delivered by the proof of the completeness theorem has a domain $|\mathfrak{A}|M$ that is no larger than the set of the terms of the language $\mathcal{L}L$. So $\mathfrak{A}M$ is at most !!denumerable. □

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